

15**A**

$$I \propto \frac{1}{r^2}$$

$$\frac{I_x}{I_{4x}} = \left(\frac{4x}{x} \right)^2 = 16$$

$$I_{4x} = \frac{1}{16} I$$

$$I \propto A^2$$

$$\text{Since } I_{4x} = \frac{1}{16} I,$$

$$\frac{I_{4x}}{I} = \left(\frac{A_{4x}}{A} \right)^2$$

$$\frac{1}{16} = \left(\frac{A_{4x}}{A} \right)^2$$

$$\frac{A_{4x}}{A} = \frac{1}{4} \quad \Rightarrow \quad A_{4x} = \frac{A}{4}$$

A**16****D****E**